Chapter 27 Review

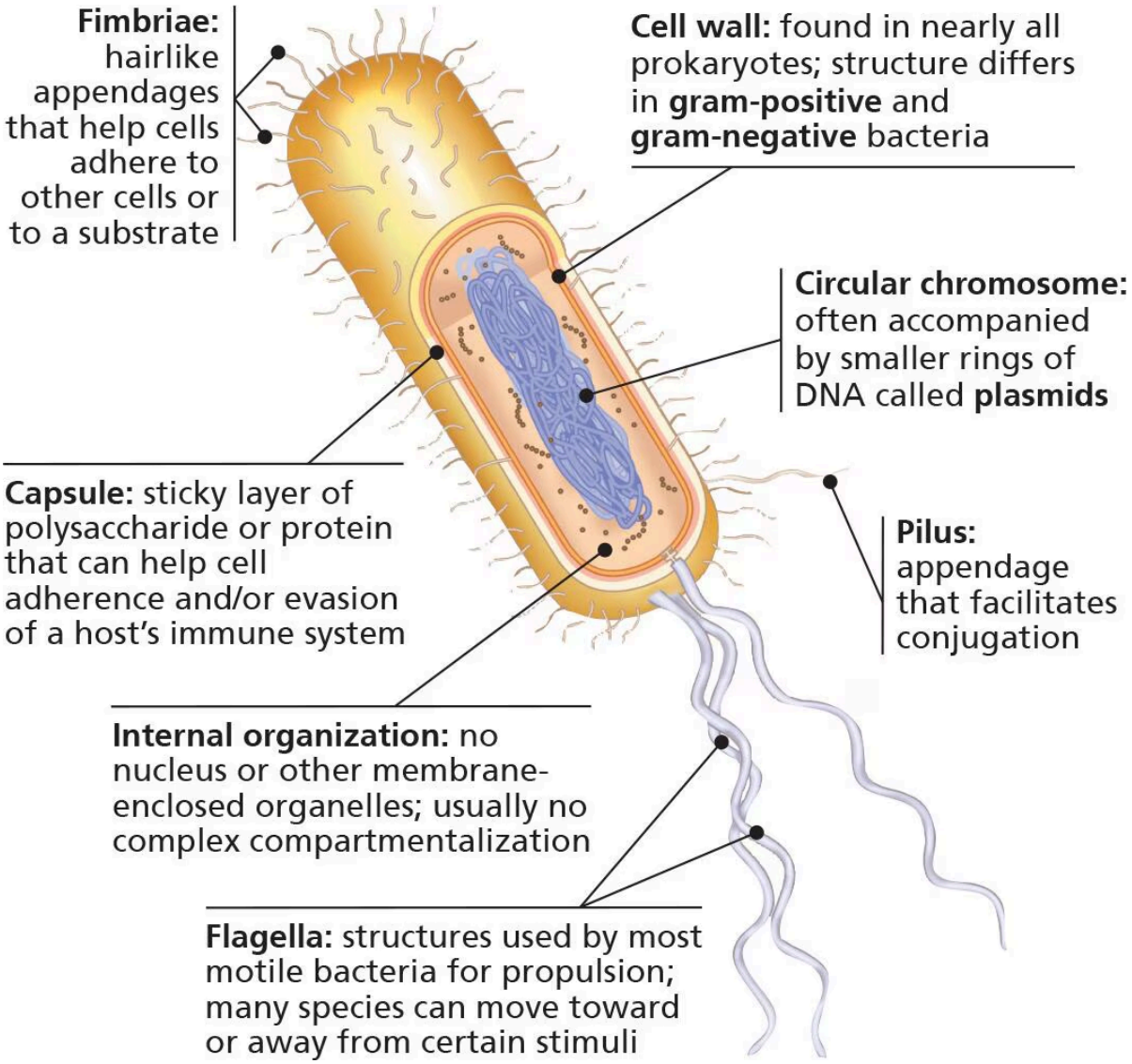
Campbell Biology, AP Edition, 12e

Chapter 27 Review

Summary of Key Concepts

Justify the claim that humans have more traits in common with the genus *Sulfolobus*, a prokaryote that lives in extremely hot and sulfur-rich water, than they do with *E. coli*, a prokaryote that lives symbiotically in human intestines. (**BIG IDEAS 1 & 4)**

Concept 27.1: Structural and functional adaptations contribute to prokaryotic success



Information is given in a table:

Structure	Function/Description
Fimbriae	Hair-like appendages on the surface of the cell that help cells adhere to other cells or substrate
Cell Wall	Found in nearly all prokaryotes; structure differs in gram-positive and gram-negative bacteria
Circular Chromosome	Often accompanied by smaller rings of DNA called plasmids.
Sex Pilus	String-like appendage (in appearance) that facilitates sexual conjugation
Capsule	Sticky layer of polysaccharide or protein that can help cell adherence and/or evasion of a host's immune system
Internal Organization	No nucleus or other membrane-enclosed organelles; usually no complex compartmentalization.
Flagella	Whip-like (in appearance) structures used by most motile bacteria or propulsion; many species can move towards or away from

- Many **prokaryotic** species can reproduce quickly by binary fission, leading to the formation of extremely large populations.

Describe features of prokaryotes that enable them to thrive in a wide range of different environments.

Concept 27.2: Rapid reproduction, mutation, and genetic recombination promote genetic diversity in prokaryotes

- Because prokaryotes can often proliferate rapidly, mutations can quickly increase a population's genetic variation. As a result, prokaryotic populations often can evolve in short periods of time in response to changing conditions.
- Genetic diversity in prokaryotes also can arise by recombination of the DNA from two different cells (via **transformation**, **transduction**, or **conjugation**). By transferring advantageous alleles, such as ones for antibiotic resistance, recombination can promote adaptive evolution in prokaryotic populations.

Mutations are rare and prokaryotes reproduce asexually, yet their populations can have high genetic diversity. Explain how this can occur.

Concept 27.3: Diverse nutritional and metabolic adaptations have evolved in prokaryotes

- Nutritional diversity is much greater in prokaryotes than in eukaryotes and includes all four modes of nutrition: photoautotrophy, chemoautotrophy, photoheterotrophy, and chemoheterotrophy.
- Among prokaryotes, **obligate aerobes** ****require O2, **obligate anaerobes** are poisoned by O2, and **facultative anaerobes** can survive with or without O2.
- Unlike eukaryotes, prokaryotes can metabolize nitrogen in many different forms. Some can convert atmospheric nitrogen to ammonia, a process called **nitrogen fixation**.
- Prokaryotic cells and even species may cooperate metabolically, including in surface-coating **biofilms**.

Describe the range of prokaryotic metabolic adaptations.

Concept 27.4: Prokaryotes have radiated into a diverse set of lineages

- Molecular systematics is helping biologists classify prokaryotes and identify new clades.
- Diverse nutritional types are scattered among the major groups of bacteria. The two largest groups are proteobacteria and gram-positive bacteria.
- Some archaea, such as **extreme thermophiles** and **extreme halophiles**, live in extreme environments. Other archaea live in moderate environments such as soils and lakes.

How have molecular data informed prokaryotic phylogeny?

Concept 27.5: Prokaryotes play crucial roles in the biosphere

- Decomposition by heterotrophic prokaryotes and the synthetic activities of autotrophic and nitrogen-fixing prokaryotes contribute to the recycling of elements in ecosystems.
- Many prokaryotes have a **symbiotic** relationship with a **host**; the relationships between prokaryotes and their hosts range from **mutualism** to **commensalism** to **parasitism**.

In what ways are prokaryotes key to the survival of many species?

Concept 27.6: Prokaryotes have both beneficial and harmful impacts on humans

- People depend on mutualistic prokaryotes, including hundreds of species that live in our intestines and help digest food.
- Pathogenic bacteria typically cause disease by releasing **exotoxins** or**** **endotoxins**. Horizontal gene transfer can spread genes associated with virulence to harmless species or strains.

- Resistance to multiple antibiotics has evolved in many pathogenic bacteria, a problem of great concern for public health.
- Prokaryotes can be used in **bioremediation** and production of plastics, vitamins, antibiotics, and other products.

Describe beneficial and harmful impacts of prokaryotes on humans.